# Introduction to Artificial Intelligence

***Simple Study Notes for the AI Study Assistant Agent***

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks that normally require human intelligence. These tasks may include understanding language, recognizing images, making predictions, solving problems, and supporting decisions.

## 1. What Is Artificial Intelligence?

AI systems use data, algorithms, and computational methods to produce useful outputs. For example, a voice assistant can interpret a spoken request, while an image classifier can identify whether a photograph contains a cat or a dog. AI does not necessarily think or understand in the same way that humans do; its abilities depend on how it is designed and trained.

## 2. Main Types of AI Approaches

| Approach | Meaning | Example |
|---|---|---|
| Machine Learning (ML) | Learns patterns from data to make predictions or decisions. | Predicting house prices from past sales. |
| Deep Learning (DL) | Uses multi-layer neural networks to learn complex patterns. | Recognizing objects in images. |
| Natural Language Processing (NLP) | Helps computers process and generate human language. | Text summarization or translation. |
| Generative AI | Creates new content such as text, images, or code. | Drafting an explanation of a science topic. |

## 3. What Is Machine Learning?

Machine Learning is a subset of AI in which a computer learns patterns from examples instead of relying only on manually written rules. A model is trained using data and then evaluated on examples it has not seen before. The quality of its predictions depends on the data, the learning method, and the way it is evaluated.

**Example:** A spam email classifier can learn from messages labelled as 'spam' or 'not spam'. After training, it predicts the likely category of a new email.

## 4. What Is a Large Language Model (LLM)?

A Large Language Model is an AI model trained on large amounts of text to generate and interpret language. Given a prompt, it predicts a sequence of tokens that forms a response. LLMs can explain concepts, summarize text, answer questions, and help draft code. However, they can sometimes produce incorrect or unsupported statements, so important answers should be checked.

## 5. What Is an AI Agent?

An AI agent is a system that works toward a goal by receiving input, deciding what step to take, and using available tools or information to produce a result. A simple agent may call a language model and use a search or calculation tool. More advanced agents can choose between tools, check results, and repeat steps when needed.

**Example:** A study assistant agent can read a student's PDF notes, retrieve relevant paragraphs, and use an LLM to answer a question using those paragraphs.

## 6. What Is Retrieval-Augmented Generation (RAG)?

RAG combines information retrieval with text generation. First, a document is split into smaller chunks. Each chunk is represented as an embedding, a numerical representation of its meaning. When a user asks a question, the system finds chunks that are semantically similar to the question. Those chunks are passed to the language model as context, helping it produce an answer grounded in the supplied documents.

**Typical RAG workflow:** PDF → text extraction → chunks → embeddings → relevant passages → LLM prompt → answer with source page references.

## 7. Benefits and Limitations of AI

**Benefits:** AI can automate repetitive tasks, process large amounts of information, support learning, and provide quick access to explanations.

**Limitations:** AI output can be inaccurate or biased, models may lack current or document-specific information, and hosted APIs may have usage limits. Human review remains important, especially for high-impact decisions.

## 8. Key Terms to Remember

- **Algorithm:** A set of steps used to solve a problem.
- **Dataset:** A collection of examples used for analysis or model training.
- **Model:** A learned or programmed system that maps inputs to outputs.
- **Prompt:** The instruction or question provided to a language model.
- **Embedding:** A vector representation that captures aspects of meaning.
- **Retrieval:** Finding information relevant to a user's query.
- **Hallucination:** An AI-generated claim that is false or not supported by evidence.

## 9. Quick Revision Questions

1. What is Artificial Intelligence?
2. How is Machine Learning related to AI?
3. What is the role of an LLM in an AI application?
4. What is an AI agent, and how can it use tools?
5. What are the main steps in a RAG pipeline?
6. Why should AI-generated answers be verified?